

# Agilent Twistorr 304 FS

## Data sheet

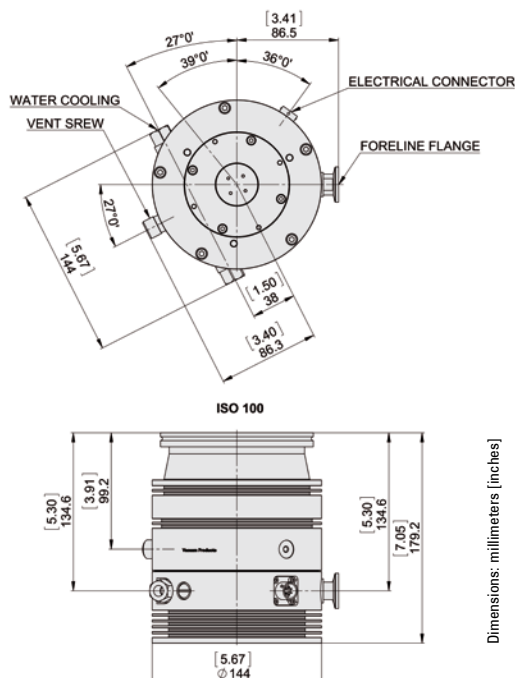


The TWISTORR 304 FS is the new generation 300 l/s Turbo Pump with Agilent Floating Suspension

● Agilent Floating Suspension, the breakthrough bearing technology that reduces

acoustical noise and vibration

- Proven best performance on the market, with new TwisTorr stages optimized for H<sub>2</sub> Compression Ratio
- Ideal for demanding instrumentation, academic and research applications



### Technical Specifications

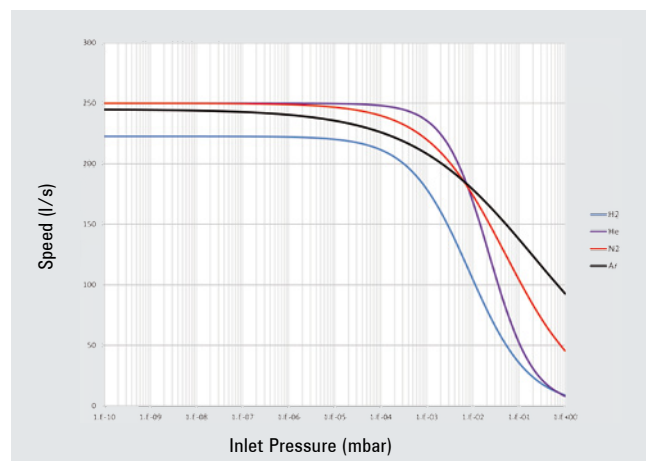
|                                                                               |                                                                                                         |                               |
|-------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|-------------------------------|
| <b>Pumping speed</b>                                                          | ISO 100 / CF 6"                                                                                         | ISO 160 / CF 8"               |
| N <sub>2</sub>                                                                | 250 l/s                                                                                                 | 250 l/s                       |
| He                                                                            | 255 l/s                                                                                                 | 255 l/s                       |
| H <sub>2</sub>                                                                | 220 l/s                                                                                                 | 220 l/s                       |
| Ar                                                                            | 250 l/s                                                                                                 | 250 l/s                       |
| <b>Gas throughput at full rotational speed</b><br>(with recommended forepump) | Ambient Temp.<br>(25°C)                                                                                 | Water Temp.<br>(25°C, 50 l/h) |
| N <sub>2</sub>                                                                | 170 sccm                                                                                                | 170 sccm                      |
| Ar                                                                            | 110 sccm                                                                                                | 110 sccm                      |
| <b>Compression ratio &amp; Foreline Tolerance</b>                             |                                                                                                         |                               |
| N <sub>2</sub>                                                                | > 1 x 10 <sup>11</sup>                                                                                  | >10 mbar                      |
| He                                                                            | > 1 x 10 <sup>8</sup>                                                                                   | >10 mbar                      |
| H <sub>2</sub>                                                                | 1.5 x 10 <sup>6</sup>                                                                                   | >4 mbar                       |
| Ar                                                                            | > 1 x 10 <sup>11</sup>                                                                                  | >10 mbar                      |
| <b>Base pressure with recommended forepump</b> (5 m <sup>3</sup> /h)          | < 1 x 10 <sup>-10</sup> mbar (< 1 x 10 <sup>-10</sup> Torr)                                             |                               |
| <b>Inlet flange</b>                                                           | CFF 8" O.D. ISO 160<br>CFF 6" O.D. ISO 100                                                              |                               |
| <b>Foreline flange</b>                                                        | KF16 NW (KF25 - optional)                                                                               |                               |
| <b>Rotational speed</b>                                                       | 60000 rpm (1010 Hz driving frequency)                                                                   |                               |
| <b>Start-up time</b>                                                          | < 3 minutes                                                                                             |                               |
| <b>Recommended forepump</b>                                                   | mechanical:                                                                                             | Agilent DS 102                |
|                                                                               | dry pump:                                                                                               | Agilent SH 110                |
| <b>Operating position</b>                                                     | Any                                                                                                     |                               |
| <b>Operating ambient temperature</b>                                          | +5 °C to +35 °C                                                                                         |                               |
| <b>Relative humidity of air</b>                                               | 0 - 90 % (not condensing)                                                                               |                               |
| <b>Bakeout temperature</b>                                                    | 80 °C at inlet flange max (ISO flange)<br>120 °C at inlet flange max (CFF flange)                       |                               |
| <b>Lubricant</b>                                                              | Permanent lubrication                                                                                   |                               |
| <b>Cooling requirements</b>                                                   | Forced air<br>(5- 35 °C ambient temperature)<br>Water (mandatory if<br>ambient temperature > 35 °C)     |                               |
| <b>Coolant water</b>                                                          | Minimum flow: 50 l/h (0.89 GPM)<br>Temperature: +15 °C to +30 °C<br>Pressure: 3 to 5 bar (45 to 75 psi) |                               |
| <b>Noise Pressure level</b>                                                   | < 50 dB(A) at 1 meter                                                                                   |                               |
| <b>Storage temperature</b>                                                    | -40° C to +70° C                                                                                        |                               |
| <b>Max altitude</b>                                                           | 3000 m                                                                                                  |                               |
| <b>Weight kg (lbs)</b>                                                        | Pump ISO 100                                                                                            | 5.5 kg (12.3)                 |
|                                                                               | Pump CFF 6"                                                                                             | 7.5 kg (16.5)                 |
|                                                                               | Pump ISO 160                                                                                            | 5.7 kg (12.6)                 |
|                                                                               | Pump CFF 8"                                                                                             | 9.7 kg (20.9)                 |



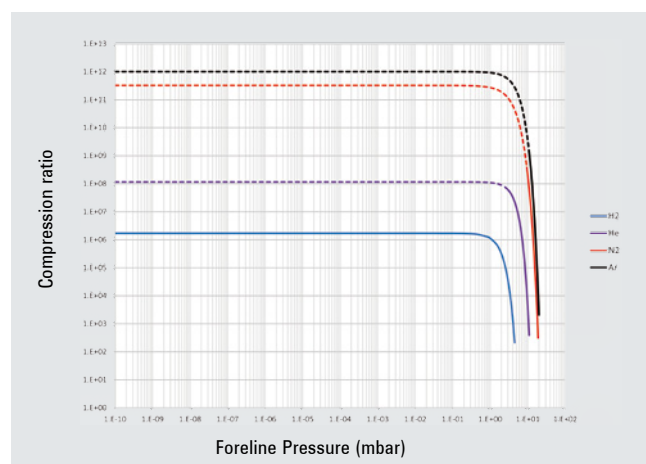
# Agilent TwisTorr 304 FS

## Data sheet

### Pumping Speed



### Compression Ratio



### Ordering Information

| Pumps                                      | Configuration Options |
|--------------------------------------------|-----------------------|
| TwisTorr 304 FS ISO100 w/ water cooling    | X3500#11#20           |
| TwisTorr 304 FS CF6" w/ water cooling      | X3500#11#21           |
| TwisTorr 304 FS ISO160 w/ water cooling    | X3500#11#22           |
| TwisTorr 304 FS CF8" w/ water cooling      | X3500#11#23           |
| TwisTorr 304 FS ISO100 w/o water cooling   | X3500#10#20           |
| TwisTorr 304 SF CF6" w/o water cooling     | X3500#10#21           |
| TwisTorr 304 FS ISO160 w/o water cooling   | X3500#10#22           |
| TwisTorr 304 FS CF8" w/o water cooling     | X3500#10#23           |
| TwisTorr 304 FS ISO100 SF w/ water cooling | X3500#11#24           |

| Controllers                                       | Part Numbers |
|---------------------------------------------------|--------------|
| TwisTorr 304 FS on board controller 24 Vdc        | X3507-64002  |
| TwisTorr 304 FS on board controller 100-240 Vac   | X3507-64003  |
| TwisTorr 304 FS AG rack controller with RS232/485 | X3506-64002  |
| TwisTorr 304 FS AG rack controller with Profibus  | X3506-64003  |

| Accessories                         | Part Numbers |
|-------------------------------------|--------------|
| Mains cable NEMA plug, 3 m long     | 9699958      |
| Mains cable European plug, 3 m long | 9699957      |
| Serial cable and T-plus Software    | 9699883      |
| Inlet Screen ISO100                 | X3500-68000  |
| Inlet Screen CF 6"                  | 9699302      |
| Inlet Screen ISO160                 | X3500-68001  |
| Inlet Screen CF 8"                  | 9699304      |

| Accessories                                     | Part Numbers            |
|-------------------------------------------------|-------------------------|
| Water cooling kit                               | 9699337                 |
| Plastic water cooling kit                       | 9699347                 |
| Air cooling kit for On board controller         | X3500-68010             |
| Fan extension cable for On board controller     | 9699949                 |
| Air cooling kit for rack AG controller          | X3500-68011             |
| Fan extension cable for rack AG controller      | 9699940                 |
| Bracket for On board controller side mounting   | X3500-68012             |
| Vibration isolator ISO 100                      | 9699344                 |
| Vibration isolator CF 6"                        | 9699334                 |
| Vibration isolator ISO 160                      | 9699345                 |
| Vibration isolator CF 8"                        | 9699335                 |
| Vibration isolator ISO 100 IDX                  | 9699396                 |
| Vent flange, NW 10 KF / M8                      | 9699108                 |
| Delay vent valve 1.2 mm orifice                 | X3505-68000             |
| Delay vent valve 0.5 mm orifice                 | X3505-68001             |
| Vent valve N.O. for rack AG controller (0.5 mm) | 9699844                 |
| Vent valve for on-board controller (1.2 mm)     | 9699834                 |
| Vent valve for on-board controller (0.5 mm)     | 9699834M006             |
| Purge valve 10 SCCM NW16KF – M12                | 9699239                 |
| Purge valve 10 SCCM ¼ Swagelok – M12            | 9699240                 |
| Purge valve 20 SCCM NW16KF - M12                | 9699241                 |
| Purge valve 20 SCCM ¼ Swagelok – M12            | 9699242                 |
| Purge valve 10 SCCM ¼ Swagelok – ¼ Swagelok     | 9699232                 |
| Purge valve 20 SCCM ¼ Swagelok – ¼ Swagelok     | 9699236                 |
| Foreline flange KF25 ¼ gas                      | 9699130                 |
| Active Gauges                                   | Ask Agilent for details |

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